Question: Simple Flower

Objective: Create a simple flower model using geometric shapes.

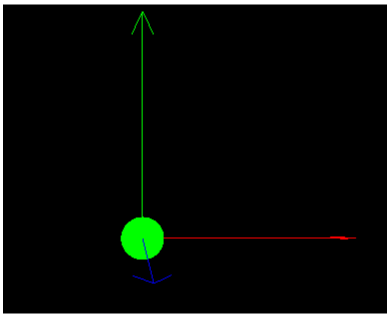
Explanation: This tests your ability to combine multiple shapes into a cohesive design, focusing on alignment and symmetry.

Description: Create a simple representation of a flower using geometric shapes and basic computer graphics tools.

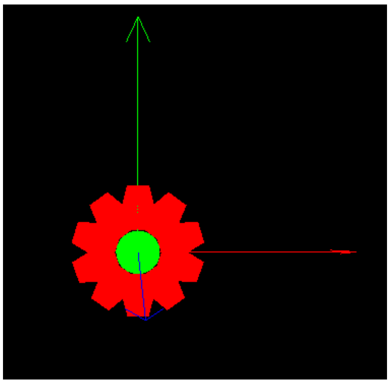
Lab Test Duration:

- Complete the lab test within the allocated time (e.g., 30 minutes).

The image below shows the given environment.



The image below shows the sample output.



Environment:

In this assignment, we will create two new files (CGLabSimpleFlower.hpp and CGLabSimpleFlower.cpp) and add them into project, please refer to Lab02 on how to create and add new files into the project.

Just as in Lab02, you will need to modify two lines in CGLabmain.cpp so that these two lines,

```
...
#include "CGLab01.hpp"
using CGLab01::MyVirtualWorld;
...
```

will be changed to...

```
...
#include "CGLabSimpleFlower.hpp"
using CGLabSimpleFlower::MyVirtualWorld;
...
```

CGLabSimpleFlower.cpp is given as below:

```
/*
 TCG6223 Computer Graphics
 Trimester 2610
 Faculty of IT, Multimedia University

 CGLabAssignment.cpp

 Objective: Lab Assignment

 Copyright (C) 2025 by Poh Li Zhe <pohlizhe@mmu.edu.my>

 This file (CGLabAssignment.cpp) can be distributed to the students

 INSTRUCTIONS
 ============
 Please refer to CGLabmain.cpp for instructions

 CHANGE LOG
 ==========
*/

#include <GL/glut.h>
#include <string>
#include <fstream>
#include "CGLabSimpleFlower.hpp"

using namespace CGLabSimpleFlower;

void SimpleFlower::drawCore(){


   glDisable(GL_CULL_FACE);
   //Quadric Object
   GLUquadricObj *pObj;
   //create and initialize quadric
   pObj = gluNewQuadric();
   glPushMatrix();
   glColor3f(0.0f, 1.0f, 0.0f);
   glTranslatef(0.0f, 0.0f, 0.0f);
   gluDisk(pObj, 0.0f, 2.0f, 26, 13);
```

```
   glPopMatrix();
   glEnable(GL_CULL_FACE);
}

void SimpleFlower::drawPetals(){
   // draw petals
}

void SimpleFlower::drawPetal(){
   // draw petal
 }
```

`CGLabSimpleFlower.hpp` is as below:

```
/*
 TCS2221 Computer Graphics
 Trimester 2, 2015/16
 Faculty of IT, Multimedia University

 CGLab01.hpp

 Objective: Header File for Lab01 Demo Models / World

 Copyright (C) 2011 by Ya-Ping Wong <ypwong@mmu.edu.my>

 This file (CGLab01.hpp) can be distributed to the students

 INSTRUCTIONS
 ============
 Please refer to CGLabmain.cpp for instructions

 SPECIAL NOTES
 =============
 * Try loading the high-polygons version of
   the Dragon model, it will be slow in slower PC.
 * Try out other models from http://graphics.stanford.edu/data/3Dscanrep/
   However, you will need to modify the file to comform to the
   format that is being used in this program.

 CHANGE LOG
 ==========
*/

#ifndef YP_CGLAB01_HPP
#define YP_CGLAB01_HPP

#include "CGLabmain.hpp"
```

```
#include <string>
#include <vector>

namespace CGLabSimpleFlower {

class SimpleFlower
{
 public:
   void drawCore();
   void drawPetal();
   void drawPetals();
};

//-----------------------------------
//the main program will call methods from this class
class MyVirtualWorld
{
 public:
   SimpleFlower     simpleFlower;
   long int timeold, timenew, elapseTime;

   void draw(){
      simpleFlower.drawCore();
      simpleFlower.drawPetals();
   }

   void tickTime()
   {
   }

   //for any one-time only initialization of the
   //   virtual world before any rendering takes place
   //   BUT after OpenGL has been initialized
   void init()
   {
   }
};

}; //end of namespace CGLab01

#endif //YP_CGLAB01_HPP
```